Ginnie Mae Project Loan CMBS Guide[1]

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Securitized Strategy

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Ginnie Mae CMBS guarantees a variety of FHA and USDA Rural Development multi-family loans that are placed into Agency CMBS pools. This guide describes the origination and guarantee process, and how investors can evaluate the prepayment risk contained in the resulting guaranteed bond classes.

From a relative value perspective the long terms of Ginnie Mae Project loans may allow them to prepay in any low rate environment caused by recession, creating an additional counter cyclical benefit beyond their government guarantee.

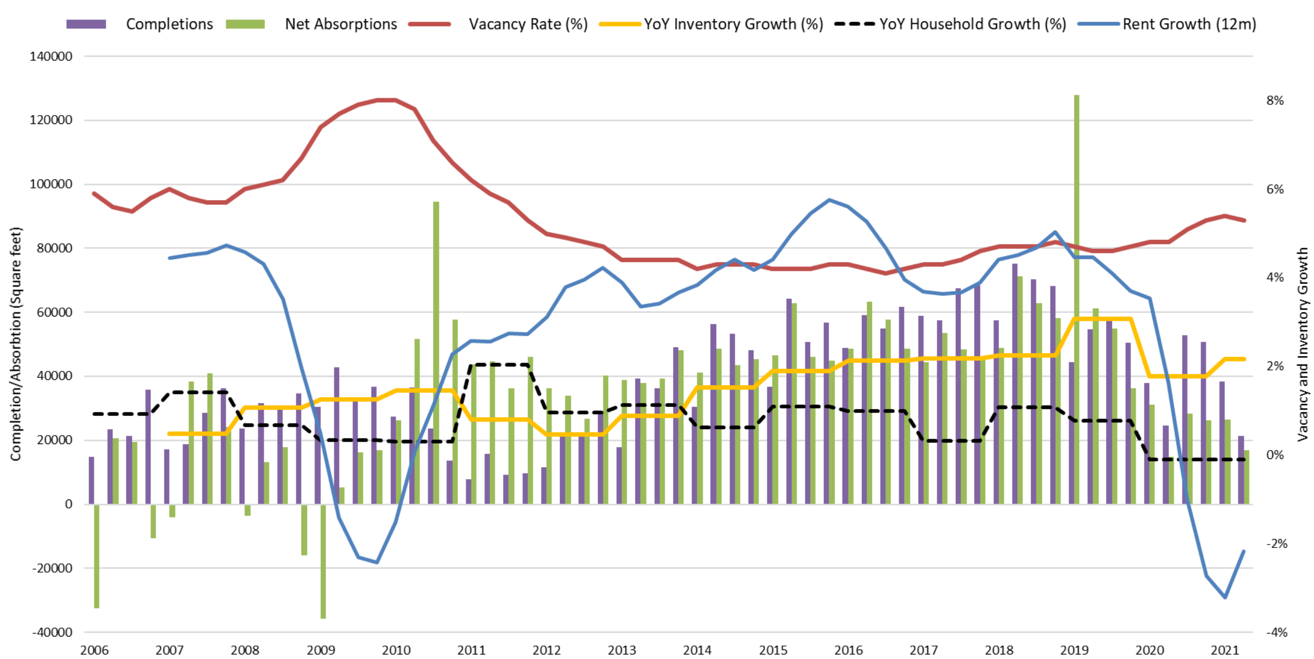
[1] This Guide updates Cantor Fitzgerald original GNR guide: “Ginnie Mae Project Loan CMBS Guide”, November 13, 2018, Darrell Wheeler.

Table of Contents Summary

1. Multifamily Fundamentals – Limited Supply Creating An Affordability Crisis

Exhibit 1 shows multifamily net completions and absorptions which have maintained a sub-6% vacancy rate since 2011. Before COVID-19, completions were exceeding absorption and creating an upward trend in vacancy, which COVID-19 helped to increase further (see the red line). This caused a dramatic drop in rental growth shown with the blue line.

Exhibit 1. Multifamily Fundamentals: Absorption, Completions, Vacancy and Rent Growth



Source: REIS, Bloomberg as of 6/30/2021.

Post-COVID tight rental markets are quickly returning.

However, in recent months there have been broad reports of decreased rental concessions and significant rental inflation, creating a consensus expectation that rents may quickly inflate (as demonstrated by the uptick in the last quarter). This rental inflation is supported by record home price appreciation, which has left many households with few accommodation alternatives. Given the sudden nature of the COVID recession this rental inflation will likely be much quicker than the previous recovery shown back in 2010.

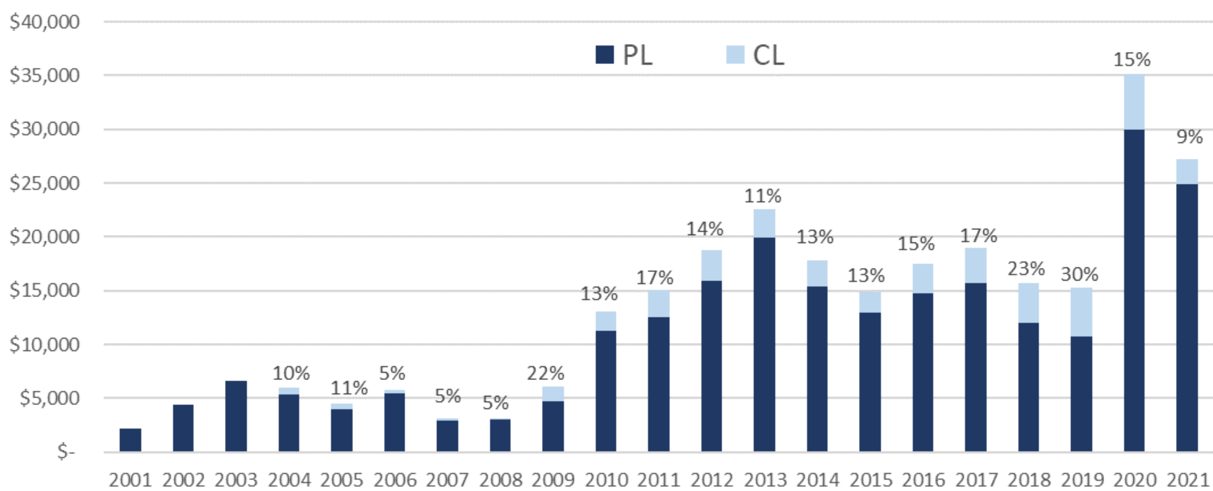
During COVID-19 many tenants missed rental payments, but low vacancy rates allowed most stabilized multifamily borrowers to continue making payments and avoid defaults. The Federal Government has been challenged to disperse COVID rental assistance via various state agencies, but eventually these payments will help tenants alleviate rental arrears and assist troubled borrowers. We will discuss this further when we review Ginnie Mae Project loan performance.

2. GNR Multifamily Issuance Grows To Fund More Construction

GNR provides a direct Federal guarantee to assist multifamily development loans and refinancing.

Governments recognize that affordable multifamily housing is necessary for a growing economy. In the United States the Federal Housing Authority and the Department of Agriculture support the financing of long term (35-40 year) multifamily mortgages for new development and stabilized properties by offering their guarantee on multifamily mortgages that meet certain housing program guidelines***. Ginnie Mae supports these Federal housing efforts by further guaranteeing (or wrapping) these 35- and 40-year housing loans (beyond the FHA and Department of Agriculture guarantee) to ensure quick payment recovery and no shortfalls.*** Exhibit 2 shows Ginnie Mae's guarantee activity with a breakdown of construction loans (CLC) and stabilized property loans (PLC).

Exhibit 2. Ginnie Mae Project Loan Activity ($MM)



Source: Intex Data Solutions

Ginnie Mae's stated mission includes "ensuring decent rental units remain accessible". They do this by lowering financing costs on a wide range of multifamily property types. ***Many of these property types (construction loans, senior housing, hospitals, or rural housing) may not have been funded without Ginnie Mae's guarantee, while the diversity of their portfolio allows them to take this risk with scale.*** The resulting bonds benefit from Ginnie Mae's guarantee of each loan's interest and principal stream along with the initial prepayment lockout period, yet any prepayment fee collections are not guaranteed. In this primer we describe the various features of these Ginnie Mae guaranteed mortgages that investors should understand in order to anticipate the related bond cashflows.

Given recent COVID economic uncertainty Ginnie Project bond investors should see a stability benefit from having strong multi-family collateral that is further supported by Ginnie's Federal loan guarantee. We next describe how Ginnie Mae's approach is unique as it supports construction funding and provides a low-cost and long-term funding option to multifamily borrowers (all three ACMBS options are summarized in Appendix A).

3. The Ginnie Mae Project Loan Process: GNR Wraps FHA Originated Multifamily Loans Creating A Range of WAL Bonds

The Government National Mortgage Association (Ginnie Mae) is part of the Department of Housing and Urban Development (HUD), which means their program is the ***most direct U.S. Government guarantee for multi-family bond issuance (and their ACMBS bonds are the only ACMBS bonds that have a 0% risk-based capital rating under Basel III). Ginnie Mae has been supporting multi-family lenders since 1972, and over the ensuing years has guaranteed >$286 billion of multi-family loans.*** Ginnie-Mae's participation is simple as they reinsure already federally guaranteed loans created by Federal Housing Administration (FHA) approved mortgage originators. ***Thus, the FHA underwrites and guarantees the loan under one of its housing policy driven programs, leaving Ginnie Mae as a second insurer with no direct involvement as an originator or as an issuer.*** A small number of these loans are also rural rental housing loans guaranteed by Rural Development under Title V of the 1949 Housing Act to facilitate housing and community facilities programs. ***The resulting loans support the full range of multi-family: low- and moderate-income multi-family, nursing homes, assisted living, condominiums, cooperatives, rural development, and even hospital and health care centers.***

Each loan is underwritten based upon the relevant section of the National Housing Act and then sized in line with the limitations of the specific referenced program section. These loan sections have been created to serve a variety of social and housing needs as listed in Exhibit 3:

Exhibit 3. Federal Housing Authority Origination Programs:

FHA Multifamily Programs	
207	Rental Housing/Manufactured Home Parks
213	Cooperative Units
220	Rental Housing for Urban Renewal and Concentrated Development Areas
221(d)(4)	New Construction or Substantial Rehabilitation of Rental Housing
207/223(f)	Purchase or Refinancing or Existing Multifamily Housing Projects
223(a)(7)	Refinancing of Existing Multifamily Housing
231	Rental Housing for the Elderly
234(d)	Mortgage Insurance for Construction or Substantial Rehabilitation of Condominium Projects
241(a)	Supplemental Loan Insurance for Multifamily Rental Housing
542(b)	Qualified Participating Entities Risk-Sharing Program
542(c)	Housing Finance Agency Risk-Sharing Program
232 and 232/223(f)	Mortgage Insurance for Nursing Homes, Intermediate Care, Board & Care and Assisted-living Facilities

Source: Federal Housing Administration

FHA loans provide long funding terms and Ginnie Mae's guarantee removes any investment credit concerns.

To provide stability to the borrower the loans usually have 35- or 40-year terms with full amortization. Loans may either be initially locked out from prepayment, start with a high fixed fee of 10% that declines 1% per year, have a steady 10% fixed fee for several years before beginning the 1% per year decline, or start at 5% and decline to 1% by the end of year 5. ***These prepayment terms are meant to discourage initial repayment, but in recent years most loans have lacked the initial lockout period, allowing many 2019 borrowers to prepay during the COVID-19 rate rally by paying the simple 10% fixed fee.*** One may have expected an 8 to 10% fee would prevent prepayments, but the recent Treasury rally and the resulting prepayments seem to prove otherwise. These loans are attractive to borrowers as they provide very long terms and still have clear prepayment flexibility without a full yield maintenance obligation.

The FHA guarantee does not cover interest beyond their debenture rate or any assignment fee.

The Agricultural loan guarantee only covers 90% of P&I.

To be included in a Ginnie-guaranteed security each mortgage note must have been endorsed by either the FHA or Rural Development using their Form RD 3565-4 note guarantee. ***These guaranteed loans are certificated into single mortgage pass-through Project Loan Certificates (PLC) or Construction Loan Certificates (CLC).*** After their respective construction period is complete, these CLC certificates convert to PLC loan certificates with the same term and rate within the REMIC. ***While the underlying loans do benefit from the FHA guarantee, in the event of a default the guarantee would not cover an assignment fee and only guarantees interest at the FHA debenture rate.*** Similarly, the rural loans are only guaranteed by the Agricultural department for 90% of interest and principal. These initial guarantees will limit losses as they give the servicer of a defaulted mortgage the option of assigning the mortgage to FHA or RD to forbear or foreclose on the mortgage. ***This means the Ginnie Mae guarantee effectively wraps any servicing costs or assignment fee with their timely interest and principal guarantee.*** **This limits the required bond analysis to how prepayments and defaults (now effectively involuntary prepayments) may affect a bond class's weighted average life.**

Before the launch of the Ginnie Mae REMIC in 2001, Ginnie Mae guaranteed loans were securitized using Fannie Mae's REMIC shelf. Since 2001 originators have pooled project loan securities under the Ginnie Mae MBS Program using the GNR shelf***. Pools typically contain mostly PLC and some portion of CLC guarantee certificates, creating diversified multi-loan pools with as many as 100 certificates in a pool.*** The issuer receives the difference between the mortgage pool interest rate and securities interest rate as a servicing fee. The issuer uses this fee to pay servicing costs, costs of managing the pool, and to pay Ginnie Mae's guarantee fee which is usually 13bp. No other fee can be charged to borrowers, but the issuer receives a servicing strip that is supposed to be 25 to 50bp.

Pre-COVID pools generally may contain as many as 30% CLC loan certificates as the FHA is financing a significant amount of new construction. ***Post COVID the percentage of CLC certificates will likely increase as the new Administration is highly focused on creating additional housing.***

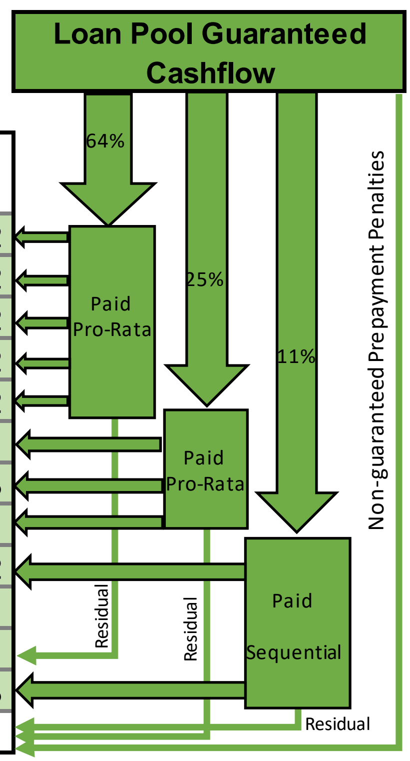
The long loan terms have created a couple different ways to allocate cashflows to bonds.

Traditionally, bond certificates have been created by pooling the cashflows creating sequential classes, an accretion class, a Z interest accrual class and an IO class. ***The accretion class and the accrual class usually have longer terms, permitting the creation of shorter term sequential classes from the 35- and 40-year cash flows.*** If a transaction does not use the accretion/accrual class structure there will be some longer bonds which are usually exchangeable with a Modifiable and Combinable Class ("MACR" class). Exhibit 4 shows this format with the guaranteed loan cashflows split into preset percentages paid to each grouping of bonds. After AB through AG are paid off any remaining cashflows are applied to the BC class which is why it has a 26.9 year weighted average life. ***The AK, BC and BL classes are exchangeable with the AM "MACR" class, blending a pass-thru class with a longer second pay sequential bond which are rarely exchanged after pricing.***

Exhibit 4. GNR 2021-22 REMIC Bond Structure

Class	Original Balance	Rate (%)	WAL (Years)	Original Maturity
AB	50,500,000	1.1	5.0	10/16/1962
AC	100,000,000	1.75	5.0	10/16/1962
AD	55,000,000	1.35	5.0	10/16/1962
AE	107,110,000	1.25	5.0	10/16/1962
AG	71,390,000	1.5	5.0	10/16/1962
AH	41,000,000	1.1	5.2	5/16/1963
AJ	85,000,000	1.35	5.2	5/16/1963
AK	21,090,000	1.5	5.2	5/16/1963
AL	65,000,000	1.35	4.9	10/16/1962
AM	25,000,000	1.5	8.5	5/16/1963
BC	3,050,000	1.5	26.9	5/16/1963
BL	860,000	1.5	25.0	5/16/1963
IO	600,000,000	0.989	5.2	5/16/1963

The AM Class can be exchanged for the AK, BC, and BL classes or vice versa.



Source: Offering Documents

The exchangeability of the AM class for these three classes allows the issuers to size the transaction structure based upon initial investor demand. The last IO class in the diagram receives any residual cashflows from the second and third cashflow stream plus any prepayment fees that are collected on the loans, but those prepayment fees are not guaranteed so it is the only class not shaded green to indicate a full guarantee.

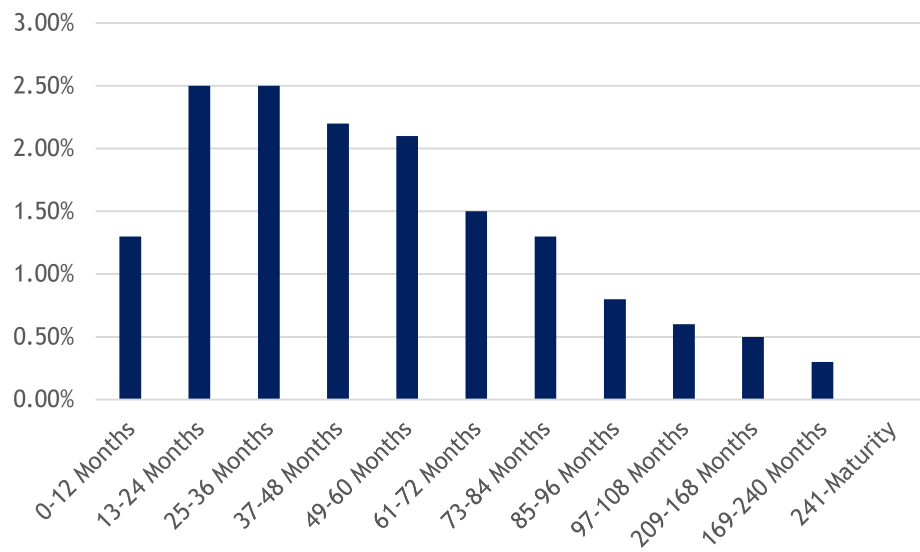
GNR pools will have bonds supported by 2 to 5 year WAL loan groupings.

Every bond structure is different as they are customized to meet investor requests, but we can generalize that each transaction cuts their cashflows to create 2 to 3 pro-rata and sequential bond groupings with different WALs. Investors should expect to see a bond grouping less than or near 5-year WALs (classes AB to AG above), another grouping beyond 5 years (class AH, AJ and AK), and then either a selection of longer classes that can be combined or the traditional accretion directed amortization class, with an accrual class and IO. We should also note that not every GNR issue uses A, B and C class designations. ***Yet many institutional investors that follow the product understand how their bond class should perform and appreciate the flexibility that can be tailored from these long-term project loans.***

4. The Long Loan Term Requires A Custom Prepayment and Default Vector:

Traditional GNR bonds have been priced with a “CPJ” prepayment vector that combines a constant annual prepayment rate with a customized constant default vector that we display in Exhibit 5.

Exhibit 5. The Project Loan CPJ Vector Includes Front-Loaded Defaults



Source: Bloomberg and Cantor Fitzgerald

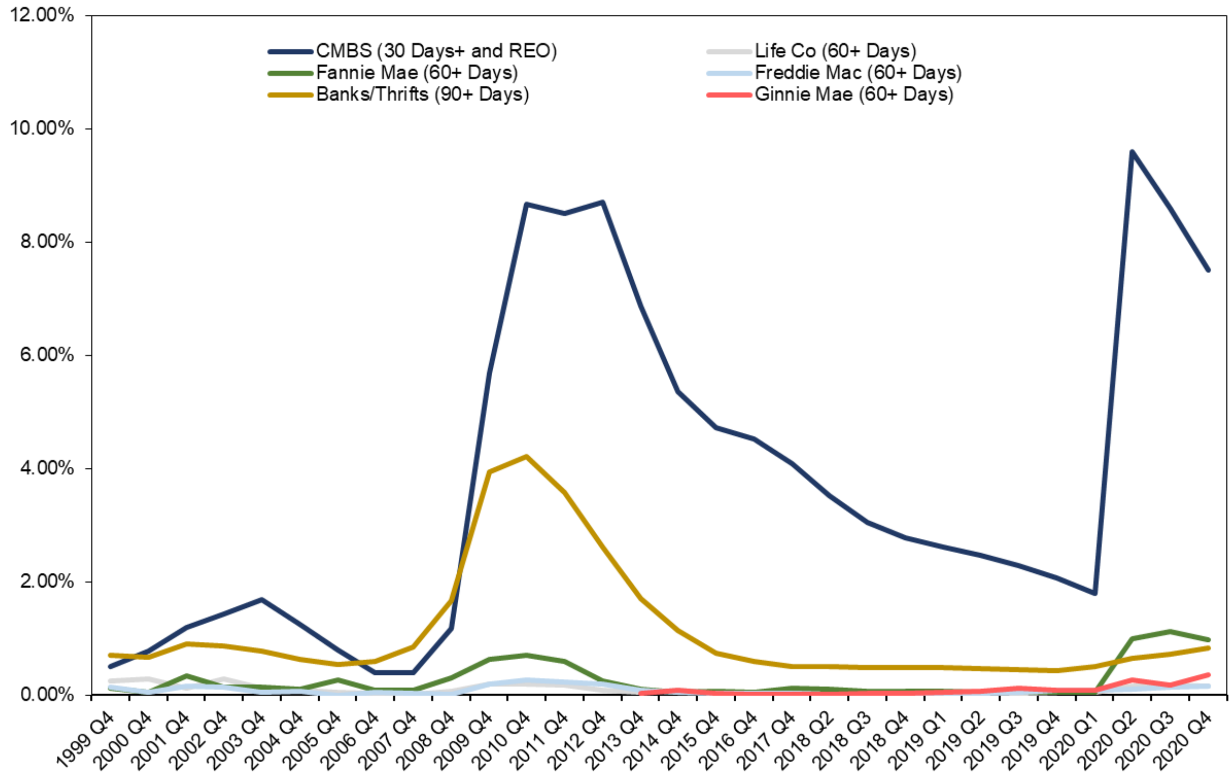
15% CPJ embeds immediate defaults, just in case we are entering a recession…

The default rates contained in CPJ are heavily front end loaded as the constant default rate starts at 1.35% and increases to 2.5% per year in year 2 and 3. ***Combining those heavy annual default rates with a 15% post lockout prepayment rate creates the 15% CPJ curve which participants appreciate for initially hitting the pool for credit risks, while allowing for prepayments that may happen once the pool has seasoned.*** This CPJ convention effectively accounts for 10% defaults in the first 60 months, which as we will discuss next is much worse than the ACMBS 2007 recession experience. This creates a worst-case pricing vector as the guaranteed par recoveries on these early defaults provide no prepayment compensation for the involuntary payments. If you compare a 15% CPR with a 15% CPJ the CPJ defaults create average lives on shorter bonds that are usually about half a year shorter, while the CPR allows the IO bond to receive more prepayment fees.

5. GNR Performance: Defaults Remained Low During COVID-19, But Prepayments Proved Sensitive To Rates

GNR defaults are rare and did not see a significant uptick even during the COVID-19 crisis. Exhibit 6 shows multifamily delinquency rates from various multifamily mortgage sources with the Ginnie Mae rate shown in red remaining below 2% throughout the pandemic.

Exhibit 6. Delinquency Rates - CMBS, Life Co, GSEs, Banks/Thrifts



Source: MBA Commercial/Multi-family Mortgage Delinquency Rates Major Investor Groups Q4 2020, Intex

Exhibit 7 breaks out these figures by vintage with higher delinquency vintages circled in red. Some of these older vintages look like they may eventually exceed 1% or even 2% of troubled loans, but that activity built up over several years and is calculated from lower pool balances. Nonetheless, the 2019 vintage now looks like it could exceed 1.5% in delinquency if the 1.3% of 30-day loans continue to transition. ***These 2019 figures are still well below the defaults contained with the CPJ projection but do demonstrate the benefit and need to have a prepayment vector that front loads some component of defaults.***

Exhibit 7. Ginnie Mae Project Loan Speeds

	Balance (MM)		WAC		CPR					Delinquency		
Issued	Curr	Orig	Curr	Orig	1M	3M	6M	12M	Life	30D	60D	90D
2001	1	2,186	8.12	7.41	0.00	0.00	0.00	12.25	27.23	-	-	-
2002	13	4,817	7.32	6.93	0.00	1.28	2.26	17.39	22.50	-	-	-
2003	73	6,256	5.88	6.05	7.47	5.08	4.83	12.30	18.74	-	-	-
2004	80	5,972	5.72	5.84	19.30	15.20	22.29	19.72	17.70	0.17	-	-
2005	129	4,481	5.48	5.66	3.14	10.01	7.06	6.20	16.72	-	0.63	-
2006	149	6,271	5.75	6.49	3.49	4.71	5.40	6.62	18.71	-	-	-
2007	117	3,931	5.96	6.36	1.00	12.06	19.74	15.01	18.95	3.03	-	0.60
2008	190	3,724	6.78	6.32	0.00	1.96	2.39	3.95	12.86	-	-	-
2009	178	5,857	5.94	5.76	0.00	0.00	4.31	9.80	21.83	1.61	-	-
2010	520	11,056	4.54	4.72	12.24	8.36	8.59	17.44	20.83	1.55	1.57	-
2011	1,536	15,166	4.08	4.43	16.87	29.37	37.05	34.70	17.16	-	0.94	0.70
2012	5,215	19,461	3.13	3.48	16.28	28.90	29.36	27.76	10.22	0.28	1.25	0.41
2013	8,413	24,016	2.99	3.19	18.89	25.20	27.06	27.96	9.15	0.19	0.16	0.22
2014	5,005	18,852	3.79	4.00	17.82	27.32	30.53	33.78	14.66	0.55	0.11	0.27
2015	5,244	15,562	3.58	3.77	21.20	29.54	34.08	35.25	13.70	-	0.27	0.01
2016	8,139	18,000	3.50	3.62	18.64	31.50	35.07	33.27	12.24	0.13	0.34	0.04
2017	9,710	18,672	3.49	3.55	36.24	32.76	34.52	34.99	12.71	0.79	-	0.12
2018	8,430	16,186	3.71	3.73	29.57	27.76	35.18	34.15	15.83	0.59	0.08	0.27
2019	7,826	14,868	3.80	3.86	29.59	29.14	34.70	32.93	21.50	1.31	0.19	0.18
2020	27,011	32,759	3.11	3.16	8.23	8.54	9.87	8.19	8.77	0.44	0.20	0.03
2021	30,314	33,480	2.78	2.79	0.98	0.50	0.76	-	1.38	0.30	0.02	0.00
Weighted Average			3.26	3.34	14.50	16.84	19.10	18.45	9.58	0.44	0.19	0.10

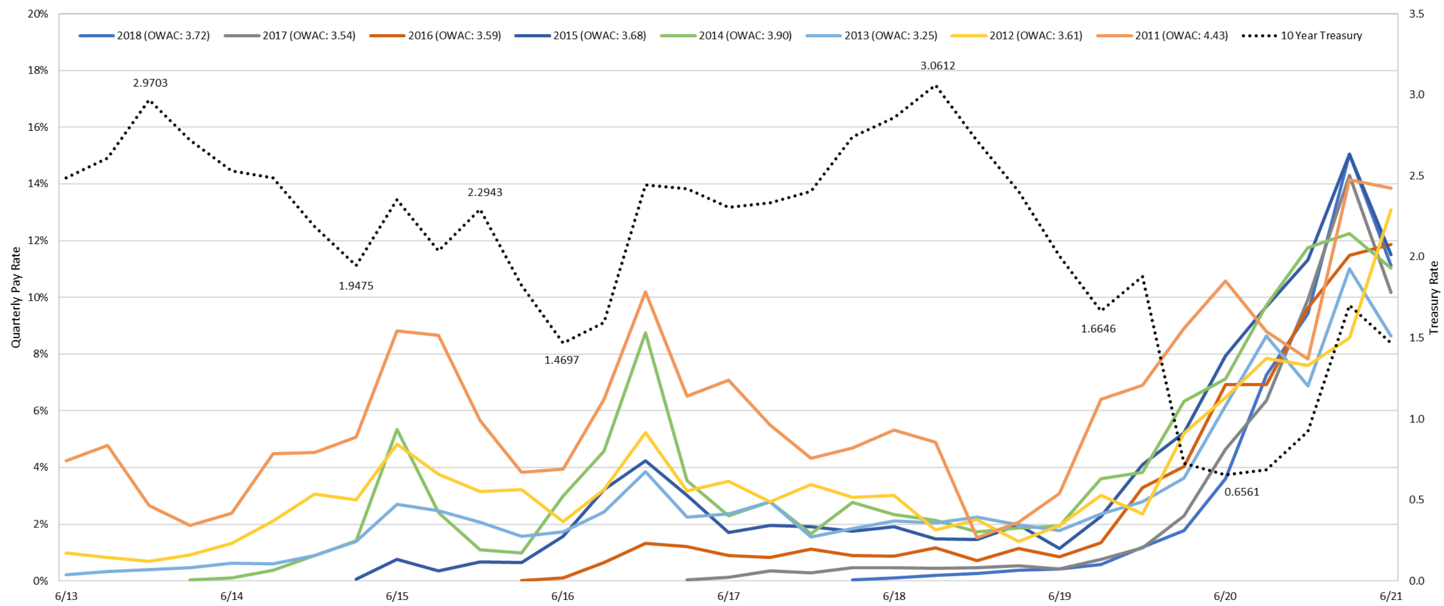
Source: Intex Data Solutions and Bloomberg, as of 8/27/2021

The 2020 COVID rate drop saw all active vintages prepay at 32-40% CPR regardless of WAC.

Exhibit 7 also shows all vintage CPRs have spiked in the past 12 months as COVID-19 quickly dropped rates 100bp, increasing the 1- to 12-month CPRs for most recent vintages from 2011 to 2019 (see green box in Exhibit 7). ***By the end of 2020 many 3- and 6-month CPRs exceeded 40% and have now recently stabilized in the low to mid-30% range***. Relative to historical GNR prepayments these recently high CPR rates are clear outliers as it usually takes several years before FHA borrowers are prepared to pay the high fixed fee to cause CPR rates to exceed 10%.

Exhibit 8 presents historical GNR prepayment rates by vintage on a quarterly basis along with the ten-year Treasury showing the last prepayment spike back in 2015 after a 2013-14 Treasury rally. ***But the early 2020 Treasury rally was double the magnitude of the 2013 rally, so every GNR vintage experienced a prepayment spike at much higher levels than the 2015 experience.*** Looking at Treasury yields shown in black there is a clear inverse relationship between Treasury yield movements and changes in prepayment speeds.

Exhibit 8. Vintage Quarterly Prepayment Rates and The 10-Year Treasury.



Source: Intex Data Solutions and Bloomberg

The legend at the top of Exhibit 8 includes the WAC for each vintage demonstrating how the prepayment spike happened regardless of their vintage coupon. These quarterly payment rates exceeded 8% and reached 12% for some vintages, translating in annual CPRs of roughly 32 to 48%. **The p*repayment spike event included the 2019 vintage that would have faced prepayment penalties of 9 or 10%, demonstrating how prepayments can increase early if the rate rally is sufficient.*** This recent experience suggests that investors should consider a range of CPR speeds when evaluating a bond.

Does Having CLC Drive CPR Relative To PLC?

The need for affordable housing should drive up the % of CLC within GNR.

2019 GNR issuance contained > 30% CLC loans, but in 2020 during COVID this dropped ~15% and has been only 9% in 2021 GNR issuance. ***As the new Administration and the FHFA take steps to increase the construction of affordable housing we expect that GNR will play a major roll and that CLC percentages will climb once again.*** This expectation begs the question of how CLC loans perform relative to stabilized PLC loans?

CLCs that are under construction contain additional risks including construction competition risk, material cost inflation and lease up risk which is why CLC loan WACs have had a 59bp average spread premium to the stabilized PLC loans over 5 recent GNR deals. ***These CLC loans***

convert to PLC loans with the same terms after construction, so the resulting higher coupon PLC loans should have a higher propensity to prepay. In Exhibit 9 we illustrate this completion-occupancy-prepay cycle by looking at changes in the underlying CLC and PLC quarterly balances from two different pools. In the first 2017 transaction the green arrows show the CL balance increasing the PL balance from the 2nd period to the 5th period on a dollar for dollar basis. The PL balance starts to pay down in 4th quarter 2019 and is interrupted by the COVID crisis. Looking at the 2019 transaction we see a similar CL to PL transfer and paydown pattern.

Exhibit 9. CLC and PLC Balance Changes - GNR 2017-81 and GNR 2019-14

GNR 2017-81		
Series	CL	PL
Original Balance	$ 10,247,031	$ 55,460,878
	Change	Change
6/30/2018	$ (1,108,638)	$ 897,138
9/30/2018	$ (1,686,527)	$ 1,469,345
12/30/2018	$ (2,928,947)	$ 2,703,739
3/30/2019	$ (2,607,853)	$ 2,372,884
6/30/2019	$ (190,514)	$ (50,206)
9/30/2019	$ 0	$ (243,206)
12/30/2019	$ (106,905)	$ (138,715)
3/30/2020	$ (1,617,647)	$ 1,217,682
6/30/2020	$ -	$ (1,026,984)
9/30/2020	$ -	$ (4,394,979)
12/30/2020	$ -	$ (21,478,822)
3/30/2021	$ -	$ (7,691,911)
6/30/2021	$ -	$ (10,501,545)

GNR 2019-14		
Series	CL	PL
Original Balance	$ 193,662,665	$ 40,403,332
	Change	Change
3/30/2019	$ (0)	$ (2,376,614)
6/30/2019	$ 0	$ (107,919)
9/30/2019	$ (15,314,197)	$ 15,172,842
12/30/2019	$ (11,874,840)	$ 11,699,392
3/30/2020	$ (46,506,317)	$ 43,184,831
6/30/2020	$ (18,057,614)	$ 17,703,579
9/30/2020	$ (32,371,459)	$ 31,927,825
12/30/2020	$ (16,810,986)	$ 5,142,464
3/30/2021	$ (19,795,211)	$ (10,513,847)
6/30/2021	$ (7,783,786)	$ (17,587,534)
7/30/2021	$ 0	$ (7,276,742)

Source Intex Data

Exhibit 9 does not track specific project loans, but in reviewing several pools we find that prepayments picked up very soon after the CLC loans converted to PLC. Much of this prepayment activity should likely be attributed to the COVID-19 Treasury rally, which caused prepayments among all recent vintage GNR (as we did see some 2016/17 deals hold back prepayment until the 2020 rate rally). ***But we also believe the higher WAC CLC loans dominate prepayments once they have stabilized their occupancy as real estate developers should be more inclined to sell stabilized properties to maximize their capital returns.*** To account for this higher prepayment inclination, investors should consider ramping up their CPR expectations when GNR pools have initial CLC concentrations higher than 20%.

GNR historical prepayments are rather unpredictable, suggesting investors should consider multiple potential outcomes.

The CPJ pricing convention which front end loads 10% of default risk is only likely to happen if the economy immediately slips into a recession. ***That type of immediate recession will also likely drop rates, allowing some of the more stable occupied properties to refinance if their borrowers are not financially stretched on other properties.*** But in a recession many borrowers may not have financial resources, suggesting CPJ speeds as slow as 4-8% CPJ are also possible. COVID-19 has seen some extraordinarily high CPR speeds with only limited defaults, which suggests a simple CPR vector of 0,15% and then 30 or 40% may also be appropriate. ***But the range of historical outcomes suggests investors should be careful not to take recent experience as an ongoing norm*** as the recent high CPR rate will likely eventually see most pools exhaust loans that can refinanced (borrower burnout). ***The key point investors should realize is that GNR loan pools with their long terms and declining fixed rate prepayment protection have the potential for almost any prepay and default scenario.***

6. GNR Sensitivity Analysis – A Range of CPJ and CPR Vectors Are used To Estimate Downside And Understand Potential GNR Bond Outcomes

Given the range of potential prepayment outcomes during the term of a GNR mortgage we consider multiple scenarios when reviewing specific GNR bonds. ***The speed scenarios used may vary with the pool coupon and the percentage of CLC loans as we expect CLC loans will be more inclined to convert and prepay in the first 48 months.*** Exhibit 10 presents this approach using the bonds from the recently priced GNR 2021-80 transaction. This transaction used an accretion directed class (V), an accrual class (Z) and an IO bond to create its WAL groupings. To consider this type of analysis we reference changes relative to the base pricing scenario bolded in the fourth column using Bloomberg values and the base 15% CPJ pricing speed.

Exhibit 10. Yield and Weighted Average Life Sensitivity Using GNR 2021-80

GNR 2021-80		8% CPJ			12% CPJ			CF Custom CPR Vector [0, 2, 5, 10, 15, 20]			Base Pricing/15% CPJ				20% CPJ		
Class	Coupon	Yield (%)	Sprd (N)	WAL (yrs)	Yield (%)	Sprd (N)	WAL (yrs)	Yield (%)	Sprd (N)	WAL (yrs)	Price	Yield (%)	Sprd (N)	WAL (yrs)	Yield (%)	Sprd (N)	WAL (yrs)
AD	1.25%	1.25%	32	5.58	1.25%	52	4.13	1.25%	32	5.60	**$ 99.92**	**1.25%**	**64**	**3.45**	1.24%	79	2.70
AL	1.50%	1.27%	35	5.58	1.20%	47	4.13	1.28%	35	5.60	**$ 101.11**	**1.14%**	**53**	**3.45**	1.04%	59	2.70
A	1.50%	1.36%	34	6.35	1.31%	50	4.72	1.35%	37	6.01	**$ 100.73**	**1.27%**	**58**	**3.94**	1.21%	68	3.07
AF	1.65%	1.42%	40	6.35	1.34%	53	4.72	1.41%	43	6.01	**$ 101.27**	**1.28%**	**59**	**3.94**	1.18%	65	3.07
AE	1.75%	1.49%	43	6.80	1.41%	55	5.04	1.48%	48	6.22	**$ 101.48**	**1.35%**	**61**	**4.20**	1.24%	67	3.27
AB	1.50%	1.43%	33	7.19	1.40%	51	5.34	1.42%	40	6.43	**$ 100.36**	**1.39%**	**61**	**4.45**	1.35%	75	3.47
AK	1.75%	1.53%	43	7.19	1.46%	56	5.34	1.51%	49	6.43	**$ 101.33**	**1.40%**	**63**	**4.45**	1.31%	70	3.47
AC	1.50%	1.50%	34	7.94	1.49%	52	5.93	1.49%	43	6.83	**$ 99.91**	**1.49%**	**64**	**4.95**	1.49%	81	3.85
AM	1.75%	1.60%	44	7.94	1.55%	58	5.93	1.58%	52	6.83	**$ 100.95**	**1.51%**	**67**	**4.95**	1.45%	77	3.85
AN	1.25%	1.40%	24	7.94	1.45%	48	5.93	1.42%	36	6.83	**$ 98.78**	**1.49%**	**64**	**4.95**	1.55%	87	3.85
AJ	1.75%	1.63%	45	8.22	1.59%	60	6.17	1.61%	53	6.99	**$ 100.76**	**1.56%**	**69**	**5.15**	1.51%	80	4.01
AG	1.25%	1.43%	25	8.24	1.49%	49	6.21	1.46%	37	7.02	**$ 98.53**	**1.53%**	**65**	**5.19**	1.60%	89	4.05
AH	1.50%	1.52%	34	8.24	1.53%	53	6.21	1.53%	44	7.02	**$ 99.70**	**1.53%**	**66**	**5.19**	1.54%	83	4.05
IO	0.91%	3.86%	266	8.22	3.21%	216	6.17	2.81%	169	6.99	**$ 8.48**	**3.06%**	**212**	**5.15**	3.38%	259	4.01
V	1.50%	1.86%	65	8.66	1.88%	70	8.23	1.89%	72	8.04	**$ 96.99**	**1.90%**	**76**	**7.76**	1.95%	89	6.81
Z	1.50%	2.20%	59	24.80	2.38%	80	19.83	2.54%	102	16.49	**$ 84.39**	**2.53%**	**101**	**16.90**	2.81%	137	13.26

Source: Bloomberg SYT.
Bloomberg-pricing value as of 8/31/21 to demonstrate sensitivity. Value is not intended to reflect current market conditions.

Faster speeds that would happen in a low rate environment suggest that GNR may have some counter cyclical benefits.

On the left side of the table we start with 8% CPJ as the slowest prepayment outcome for the bonds. ***Given recent prepayment speeds this 8% CPJ scenario is viewed as unlikely, yet is possible if interest rates increase from recent levels.*** The second scenario presents a slightly faster 12% CPJ. ***Both of these slower CPJ scenarios create lower spreads and yields for many of the bond classes and would usually be the worst-case scenario for shorter bonds (circled in green in the table). Yet the spreads are still 24bp to mid-70s on this guaranteed paper.*** The third scenario presents a custom scenario that starts out slow and then ramps up only CPR, just in case rates do fall further (0% for 12 months, 15% for 12 months, and then 30% CPR thereafter). Finally, we have 20% CPJ which also seems possible if a 2022 recession was to drive down interest rates, as that would once again permit certain well occupied multifamily loans to quickly prepay. ***These higher prepay scenarios may take time to develop, making them a focus for longer WAL bonds (circled in yellow). This faster speed scenario also reflects possible lower rates created by a recession and yet would significantly boost most bond yields and spreads (to be 60 - 140bp).*** Along with that observation we would note that the V classes have been structured to be fairly stable under extension or slower speed scenarios, and they show a significant spread pick up at the 20% CPJ speed.

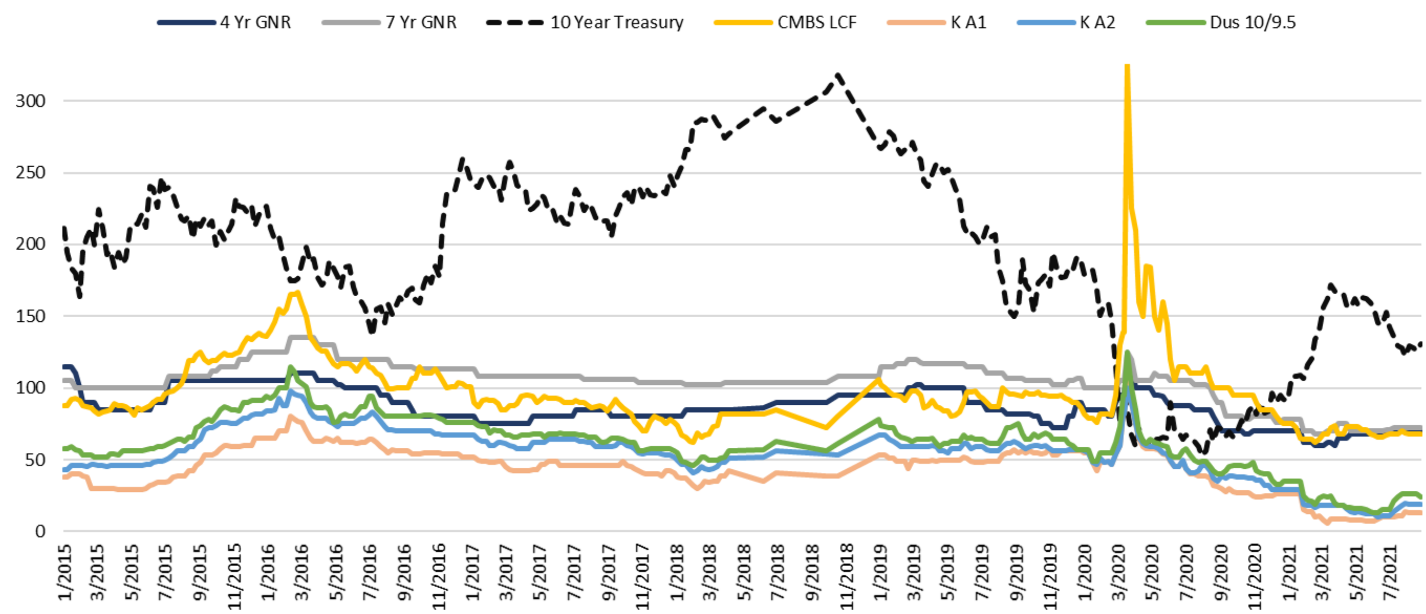
Ginnie bond classes price with significant spread to account for the WAL volatility that comes with the longer loans.

The range of spread outcomes from 24 to 137 bps highlight how project loans have considerable variability, and yet the slow speed worst case bond return is still much better than a Freddie K A1 spread. These results beg the question "what is the right spread premium for each bond class", especially when faster speeds could increase spreads on even the shorter classes by an additional 24 to 47 bps as CPJ speeds increase from 8% to 20%. ***This sensitivity analysis suggests there could be a counter intuitive return benefit for GNR in a lower rate environment.*** For this sensitivity analysis investors should consider the underlying mortgage coupons, seasoning and potential loan performance while remaining open to a full range of prepayment expectations as slower speeds will also decrease returns. ***Basically, the unknown prepayment aspect of GNR creates the spread premium to other ACMBS and requires that pools be evaluated with an appropriate worst case speed assumption along with an upside scenario.***

7. Ginnie Mae Relative Value – Guaranteed Agency Bonds + a WAL Variability Premium

The previous section mentioned Freddie K and Fannie DUS offer tighter spreads due to their tighter principal window and more certain WAL. In Exhibit 11 we show that this certainty allows those two bond sectors to price at very tight spreads relative to the 10-year Treasury and in relation to CMBS and GNR spreads.

Exhibit 11. GNR Spreads Versus Freddie K, Fannie DUS and Private CMBS LCF



Source: Cantor Fitzgerald and Bloomberg

Short and long Ginnie spreads are now more than 50bp back of other Agency CMBS.

Exhibit 11 shows fairly steady ACMBS spreads until March of 2020 when they ratchet to new tight levels as COVID-19 shut down the economy and as the Federal government implemented active bond programs for ACMBS. ***The GNR series in Exhibit 11 benefited from the Federal Reserve support, but that rally somewhat reflects market prepayment expectations that also firmed as the 10-year Treasury rallied.*** Pre-COVID19 GNR market participants were pricing in a broader range of prepayment scenarios for GNR bonds, but as the Treasury rally narrowed potential outcomes the GNR spread also rallied. In fact, the greater certainty created post COVID-19 has allowed the 15% CPJ based spread in Exhibit 11 to come in to be 60-80bp, as opposed to the 100bp+ levels that existed just before COVID-19.

Nonetheless, if we look at the GNR 4-year spread of 70bp and run prepayment scenarios that range from 8% CPJ up to 20% CPJ, prepayment adjusted spread exceeds the Freddie K A1 spread by at least 10bps. ***Overall prepayment adjusted GNR spreads usually contain a***

At worst case speed GNR has 15bp of excess spread relative to Freddie K and DUS.

15-30bp basis to Freddie K A1 bonds of similar WAL. This is an attractive spread pickup on government guaranteed debt, but the GNR market does have somewhat less liquidity than the Freddie K or Fannie Mae DUS market. ***Similarly, the recent 7-year Ginnie spread of ~72bp at 15% CPJ may have 15 to 30bp of prepayment downside at the slower speeds shown in our previous sensitivity analysis.*** That suggests the 7-year Ginnie paper would still have ~30bp of prepay adjusted spread pick up to Freddie K and DUS of similar WAL.

Beyond the comparison with other Agency bonds, Ginnie investors should also consider a comparison with CMBS last cash flow bonds and other sectors. Exhibit 12 shows that the 4-year Ginnie Mae class is now 2bp wider than the 10-year last cash flow CMBS super-senior LCF spread (the yellow line). This suggests Ginnie Mae 4-year spreads are cheap to CMBS as historically this spread has seen 4-year GNR spreads 12bp inside of CMBS LCF. We highlight this misalignment as the graph shows the CMBS LCF bond is more volatile than GNR spreads, as it bumped out to 149bp in early 2016 and then to 325bp during COVID-19. ***It is typically during those crisis periods when the stability benefit of buying Ginnie CMBS paper becomes most apparent, not to mention the benefit of the guarantee. Overall, we expect that GNR spreads could rally > 10bp based upon their recent CMBS differential.***

Exhibit 12. Agency CMBS Spreads Pre-COVID to Now (January 2020 - present)

	1/3/20	2/7/20	3/6/20	4/3/20	5/7/20	6/5/20	7/2/20	8/7/20	9/4/20	10/2/20	11/2/20	12/4/20	1/7/21	2/5/21	3/5/21	4/1/21	5/7/21	6/4/21	7/2/21	8/27/21
CMBS Fixed-Rate																				
A4 LCF	82	82	130	210	150	120	115	105	100	95	98	85	75	64	68	68	73	67	68	68
Fannie Mae																				
FNMA GeMS A1	58	50	60	85	64	50	48	46	38	32	28	27	27	14	13	6	6	6	7	14
7/6.5 DUS	50	47	67	75	59	44	48	42	33	36	32	25	25	11	9	11	8	8	11	15
10yr FNMA GeMS A2	57	53	67	85	64	52	50	48	40	40	40	33	32	20	20	20	16	13	12	22
10/9.5 DUS	57	55	73	85	62	53	58	49	40	46	48	36	35	22	25	18	16	13	15	24
Fannie SARM (DM)	67	59	66	105	80	60	59	48	35	37	35	35	40	24	25	26	24	23	22	22
Freddie Mac K																				
5 Yr K A1	55	48	60	75	58	49	44	39	30	27	25	25	26	14	8	9	8	7	10	13
10 Yr K A2	56	48	60	75	59	50	44	46	39	38	37	29	29	18	18	18	14	12	11	19
10 Yr K AM	62	53	65	80	60	55	50	53	44	43	42	35	34	23	23	25	19	17	16	27
Ginnie Mae																				
4 Yr GNR	85	85	95	115	95	88	88	85	70	70	70	70	70	62	60	65	68	68	69	70
5 Yr GNR	92	88	82	76	76	76	75	75	75	75	75	75	74	71	64	71	69	69	70	71
7 Yr GNR	100	100	105	120	110	105	105	102	90	80	80	80	78	70	68	75	70	70	71	72

Source: Bloomberg, Cantor Fitzgerald

The market is pricing Ginnie bonds at worst case slow prepayment rates, but ongoing low interest rates could deliver excess returns.

In fact if there is an economic slowdown the lower rate environment will drive prepayments with fees while Ginnie Mae's guarantee will protect bondholders from price volatility. The current 10-30bp plus prepayment adjusted differential with similar WAL Freddie and Fannie bonds suggests that the market is over-anticipating how much slowing prepayments could erode the Ginnie Mae bonds' spread advantage.

Appendix A. Summary - CMBS Multi-Family Bond Exposures

Product	GNR	Fannie Mae DUS	Freddie Mac K	Conduit CMBS
2021 Issuance/ Outstanding	**$31.2B/$118.4 billion**	**$34.3B / $341.82B**	**$42.7B/ $314.9B**	**$17.3B/382.6B (all property types)**
Collateral Description	Multi-family or senior housing loans originated by a Fair Housing Act "FHA" approved lender that are insured by FHA or the Department of Agriculture Rural Housing Service. Loans have 35 or 40 year terms with full amortization and are on either stabilized multi-family properties (Project loan Commitments "PLC") or Construction Loan Commitments (CLC).	30 approved Delegated Underwriters & Servicers create individual property loan pass through certificates that settle after 30 to 45 days (DUS MBS). DUS lenders originate the loans without Fannie Mae preapproval as they service the loan and share in its first loss. Loans are prepayable usually with a yield maintenance penalty.	20 Seller/Servicers create $5-100 million nonrecourse loans that are subject to Freddie credit approval. Loans usually have initial 2 year lockout followed by defeasance period ending 3 months prior to the maturity date. Loans can incur future subordinate debt subject to DSCR and LTV tests. Floating-rate loans usually have 1 year lockout and 1% prepayment penalty.	Fixed-rate Multi-loan CMBS pools usually contain 10 - 30% multi-family exposure, which diversifies the pool's property type specific risk. Most loans are 10 year fixed-rate loans with a 9.75 year prepayment protection from defeasance or yield maintenance.
Bond Structure	Ginnie Mae approved sponsors buy and pool FHA insured loans from originator/ issuers. Ginnie Mae guarantees timely payment of principal and interest. While collateral is locked out and has prepayment fees, there is no prepayment guarantee. Bond Classes are typically time tranched, have an IO and many other planned classes or even a Z interest accrual class.	GeMS deals are assembled from pooling the various pass through certificates and usually have a shorter and a longer class. Fannie guarantees full and timely payment of interest and principal. No guarantee on prepayment penalties, and loan yield maintenance may differ from required bond yield maintenance due to coupon basis and a final six month loan open period.	Freddie Mac guarantee on two senior time tranched classes, and AM and some IO classes. Guarantee covers timely interest payment, payment of principal at the maturity date of each loan, reimbursement of any realized losses and certain trust expenses and ultimate payment of principal by the assumed final distribution date. Current structure uses a single non-guaranteed first loss class that is 5% of pool or 10% for FRESB or KF transactions.	All properties types are pooled, which dilutes multi-family property exposure. With no guarantee, losses are applied sequentially starting with the lowest credit class, while prepayments are applied to the senior shortest WAL *pari passu* triple-A rated classes and then sequentially down into the bond structure.
Loan Recovery Process	If a borrower misses a payment the issuer undertakes foreclosure and appropriate action which includes the collection of insurance and guaranty benefits. If this is insufficient to make the mortgage P&I payments then Ginnie Mae is responsible for the shortfall.	Fannie Mae guarantee of timely interest and principal causes them to remove loans from the pool for workout recovery while continuing to make the scheduled interest and principal payments until loan was scheduled to be open to prepayment. Originator shares loss with Fannie Mae.	Defaulted loan worked out within the trust subject to pooling and servicing agreement. Originators do not share in loss. If a defaulted loan reaches its maturity the principal guarantee pays principal to the guaranteed classes.	Loans are worked out within the trust by the special servicer subject to the pooling and servicing agreement.
Rating Agency Assessment	No rating required.	Given full guarantee of timely interest and principal, no rating required..	Senior guarantee levels are rated AAA and junior AM guarantee class can also receive a non-guarantee rating. Current single nonguaranteed class is not rated as it is a first loss.	Levels justified by collateral leverage and credit enhancement.
Typical Ticker & Series Naming Convention	GNR-20yy-xx	FNA 20yy-Mxx	FREMF 20yy + - K5xx: 5 yr fixed-rate loans K7xx: 7 yr fixed-rate loans K0xx: 10 yr fixed-rate loans K15xx: 12-15 yr fixed-rate KFxx: Floating-Rate Loans	Program 20yy-Series
Information Sources	*Ginniemae.gov* or *Structuredginniemaes.ginnienet.com/multifam/*	*Fanniemae.com/port/funding-the-market/multi-family*	*https://mf.freddiemac.com/?gclid=EAIaIQobChMIwsS2ufOr8QIVwwMGAB13xAFHEAAYASAAEgKYgvD_BwE*	
Risk Based Capital	0% RBC	20% RBC	20% RBC on guaranteed Classes.	20% on AAA Super Senior Classes.

Source: Fannie Mae, Freddie Mac, and Ginnie Mae websites, Cantor Fitzgerald

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